MAHATMA GANDHI UNIVERSITY, KOTTAYAM

FIRST SEMESTER MGU-UGP (HONOURS) REGULAR EXAMINATION NOVEMBER 2024
First Semester

Multi-Disciplinary Course - MG1MDCCSC101 - INTERNET AND WEB TECHNOLOGIES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour Maximum Marks: 35

Scheme for Evaluation

Part A: Very Short Answer Questions (Answer all) - (10 * 1 = 10 Marks)

1. **Recall three devices commonly used in computer networks.** [K] [1]

- **Answer:** Any three devices (1 mark for all three, 0.5 marks for any two correct devices)
    - Router
    - Switch
    - Hub
    - Network Interface Card (NIC)
    - Modem

2. **Distinguish between microwaves and radio waves.** [U] [1]

- **Answer:** (1 mark for any one valid difference)
    - Microwaves have higher frequencies and shorter wavelengths than radio waves.
    - Microwaves are used for point-to-point communication, while radio waves can be used for broadcasting over a wider area.

3. **Define IP address.** [K] [1]

- **Answer:** A unique numerical address assigned to every device connected to a network. (1 mark)

4. **Differentiate World Wide Web and Internet.** [U] [1]

- **Answer:** (1 mark for any one valid difference)
    - The Internet is a global network of interconnected computers and servers. The World Wide Web is a collection of interconnected documents and resources, accessed over the Internet.
    - The Internet is the infrastructure, while the WWW is a service built on top of it.

5. **Name two advantages of using the internet.** [U] [1]

- **Answer:** Any two advantages (0.5 marks each)
    - Access to information
    - Communication
    - E-commerce
    - Entertainment
    - Education

6. **List the categories of firewalls.** [K] [1]

- **Answer:** Any two categories (.5 mark each)
    - Software Based Firewall
    - Hardware Based Firewall

7. **Identify the tag used for headings in HTML.** [U] [2]

- **Answer:** `<h1>` to `<h6>` (1 mark for any correct heading tag)

8. **How can you align text to the left, right, or center?** [U] [2]

- **Answer:** Using the `align` attribute within paragraph tags (`<p>`) or using CSS (1 mark for either method)

9. **Identify the HTML tag used to create a marquee effect.** [U] [2]

- **Answer:** `<marquee>` (1 mark)

10. **Name any two attributes in a `<form>` element** [K] [2]
    - **Answer:** Any two attributes (0.5 marks each)
        - `action`
        - `method`
        - `enctype`
        - `target`

Part B: Short Answer Questions (Answer any 3 out of 5 Questions) - (3 * 5 = 15 Marks)

11. **Compare LAN and WAN in terms of coverage and speed.** [U] [1]
    - **Answer:** (2.5 marks for comparison in terms of each factor)
        - **Coverage:** LANs cover smaller geographical areas (homes, offices, buildings), while WANs cover larger areas (cities, countries, or even globally).
        - **Speed:** LANs generally offer higher speeds than WANs.
12. **Discuss the importance of security in a network.** [U] [1]
    - **Answer:** (1 mark for each aspects like confidentiality, integrity, availability, and threats/vulnerabilities)
13. **Differentiate between static and dynamic web pages** [U] [2]
    - **Answer:** (2.5 marks for explaining each type with key differences)
        - **Static:** Content remains the same for every user.

        - **Dynamic:** Content can change based on user interaction, database information, or other factors.
14. **Explain the RGB color model. How are colors represented using RGB values? Provide an example of an RGB color code.** [U] [2]
    - **Answer:** (2 marks for explaining the model, 2 marks for explaining how colors are represented, 1 mark for a valid example)
        - RGB uses varying intensities of red, green, and blue light to create all colors.
        - Each color is represented by three values, typically ranging from 0 to 255.
15. **Describe the syntax of creating a hyperlink in HTML.** [U] [2]
    - **Answer:** `<a href="url">Link text</a>` (2 marks for the correct syntax and 3 marks for explanation of the elements)

Part C: Essay Questions (Answer any 1 out of 2 Questions) - (1 * 10 = 10 Marks)

16. **Examine the various types of network topologies.** [U] [1]
    - **Answer:** (1 marks each for explaining topology and one mark each for the corresponding diagram)
        - Bus, star, ring, mesh, tree
17. **Discuss the common HTML tags used for basic text formatting, such as bold, italic, underline, and strikethrough. How are these tags used to emphasize or modify text? Provide examples of their usage.** [U] [2]
    - **Answer:** (2.5 mark for each tag with explanation and example)
        - `<b>` or `<strong>` for bold
        - `<i>` or `<em>` for italic
        - `<u>` for underline
        - `<s>` or `<strike>` or `<del>` for strikethrough